

matplotlib.pyplot.plot()

A Compact Reference

A NumPy `axis` primer, then line styles, markers, colors, format strings, and arguments

The NumPy `axis` argument (quick primer)

An array's *axes* are its dimensions. For a 2-D array of shape (`rows`, `cols`), `axis=0` runs *down* the rows and `axis=1` runs *across* the columns. Reductions such as `sum`, `mean`, `max`, and `std` take an `axis` telling them which dimension to work along.

Rule of thumb: the axis you *name* is the axis that **disappears**. `axis=None` (the default) reduces the whole array to a scalar; `keepdims=True` keeps the collapsed axis as length 1.

`axis=1` $\longrightarrow$
`axis=0` $\downarrow$ $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$
 shape (2, 3)

```
A = np.array([[1,2,3],[4,5,6]])
A.sum()          -> 21          # everything
A.sum(axis=0)    -> [5, 7, 9]  # per column
A.sum(axis=1)    -> [6, 15]    # per row
```

Plotting tie-in: `plt.plot(A)` on a 2-D array draws each column as its own line.

Line styles (linestyle, or inside `fmt`)

Style	String	Appearance
Solid	'_'	—————
Dashed	'--'	- - - - -
Dash-dot	'-.'	- . - . - .
Dotted	':'	
None	'' or 'None'	no line

Marker styles (marker, or inside `fmt`)

Marker	Description
'.'	Point
','	Pixel
'o'	Circle
'v'	Triangle down
'^'	Triangle up
'<'	Triangle left
'>'	Triangle right
'1'	Tri-down
'2'	Tri-up
'3'	Tri-left
'4'	Tri-right
's'	Square

Marker	Description
'p'	Pentagon
'*'	Star
'h'	Hexagon 1
'H'	Hexagon 2
'+'	Plus
'x'	X (thin)
'X'	Filled X
'D'	Diamond
'd'	Thin diamond
' '	Vertical line
'_'	Horizontal line

(last two were garbled in the source)

Colors

Single-letter codes

Code	Color
'b'	Blue
'g'	Green
'r'	Red
'c'	Cyan
'm'	Magenta
'y'	Yellow
'k'	Black
'w'	White

Default color cycle

Code	Color	Code	Color
'C0'	Blue	'C5'	Brown
'C1'	Orange	'C6'	Pink
'C2'	Green	'C7'	Gray
'C3'	Red	'C8'	Olive
'C4'	Purple	'C9'	Cyan

fmt string examples (color + marker + line style)

fmt	Meaning
'r-'	Red solid line
'bo'	Blue circles (no line)
'g--'	Green dashed line
'k:'	Black dotted line
'm-.'	Magenta dash-dot line
'ro-'	Red solid line with circle markers
'bs--'	Blue dashed line with square markers
'g^:'	Green dotted line with triangle-up markers
'C2o-'	Default green (C2), circles, solid line

Other commonly used arguments

Argument	Purpose	Example
color	Line / marker color	color='red'
linestyle	Line style	linestyle='--'
linewidth	Line thickness	linewidth=2
marker	Marker type	marker='o'
markersize	Marker size	markersize=8
label	Legend text	label='Signal'
alpha	Transparency (0–1)	alpha=0.6

Most commonly used combinations

Plot type	Example
Solid line	plt.plot(x, y, '-')
Dashed line	plt.plot(x, y, '--')
Circle markers	plt.plot(x, y, 'o')
Line + circles	plt.plot(x, y, 'o-')
Line + squares	plt.plot(x, y, 's-')
Line + triangles	plt.plot(x, y, '^-')
Dotted line + circles	plt.plot(x, y, 'o:')
Dashed line + stars	plt.plot(x, y, '*--')

The handful you will actually use

For most signal-processing and scientific plotting:

- **Line styles:** `'-'`, `'--'`, `'.'`, `'-.'`
- **Markers:** `'o'`, `'s'`, `'^'`, `'*'`, `'x'`, `'+'`, `'.'`
- **Colors:** `'C0'–'C9'` (recommended) or `'b'`, `'r'`, `'g'`, `'k'`